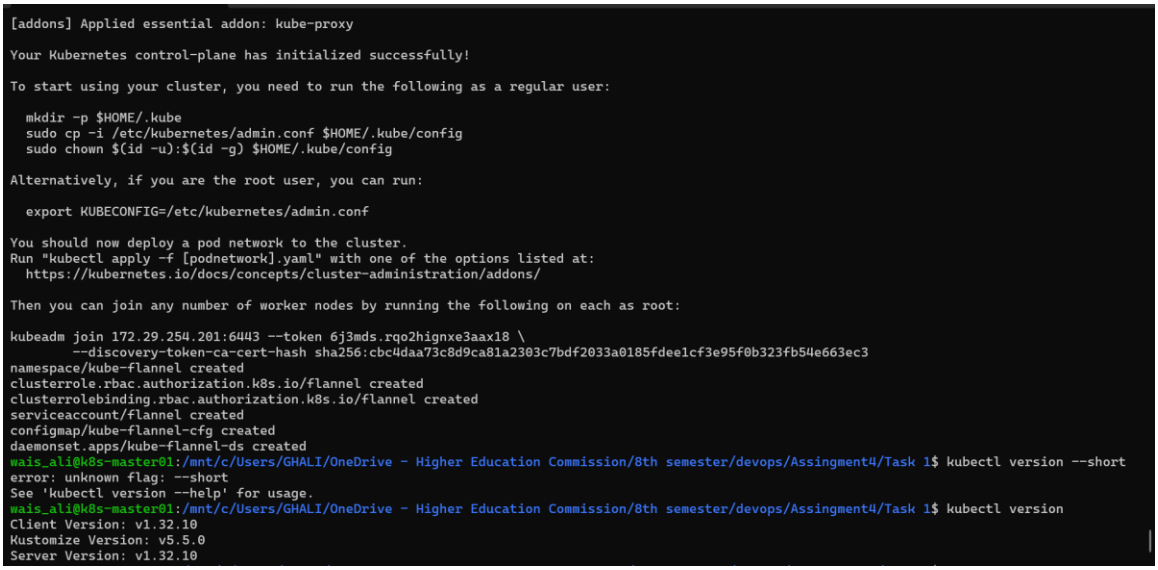
TASK# 1

1. Updates the APT package index and upgrades all installed packages.
   sudo apt update && sudo apt upgrade -y
2. Reboots the system to apply kernel updates.
   sudo reboot
3. Disables swap memory temporarily.
   sudo swapoff -a
4. Disables swap permanently by editing the fstab file.
   sudo sed -i '/ swap / s/^/#/' /etc/fstab
5. Verifies that swap is disabled.
   swapon --show
6. Loads the overlay filesystem kernel module.
   sudo modprobe overlay
7. Loads the bridge netfilter kernel module.
   sudo modprobe br_netfilter
8. Configures kernel modules to load automatically on boot.
   sudo tee /etc/modules-load.d/k8s.conf << EOF
   overlay
   br_netfilter
   EOF
9. Configures required Kubernetes networking kernel parameters.
   sudo tee /etc/sysctl.d/k8s.conf << EOF
   net.bridge.bridge-nf-call-iptables = 1
   net.bridge.bridge-nf-call-ip6tables = 1
   net.ipv4.ip_forward = 1
   EOF
10. Applies all kernel parameter changes.
    sudo sysctl --system
11. Installs the containerd container runtime.
    sudo apt install -y containerd
12. Creates the containerd configuration directory.
    sudo mkdir -p /etc/containerd
13. Generates the default containerd configuration file.
    sudo containerd config default | sudo tee /etc/containerd/config.toml
14. Configures containerd to use systemd cgroups.
    sudo sed -i 's/SystemdCgroup = false/SystemdCgroup = true/'
    /etc/containerd/config.toml
15. Restarts the containerd service.
    sudo systemctl restart containerd
16. Enables containerd to start on system boot.
    sudo systemctl enable containerd

17. Updates the package index before adding Kubernetes repository.
    sudo apt-get update
18. Installs required packages for secure repository access.
    sudo apt-get install -y apt-transport-https ca-certificates curl gpg
19. Creates a directory for Kubernetes APT keyrings.
    sudo mkdir -p /etc/apt/keyrings
20. Downloads and installs the Kubernetes repository signing key.
    curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.32/deb/Release.key | sudo gpg --dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg
21. Adds the Kubernetes v1.32 APT repository.
    echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.32/deb/ /' | sudo tee /etc/apt/sources.list.d/kubernetes.list
22. Refreshes the package index after adding Kubernetes repository.
    sudo apt-get update
23. Installs Kubernetes components kubelet, kubeadm, and kubectl.
    sudo apt-get install -y kubelet=1.32.* kubeadm=1.32.* kubectl=1.32.*
24. Prevents Kubernetes packages from being automatically upgraded.
    sudo apt-mark hold kubelet kubeadm kubectl
25. Sets a unique hostname for the Kubernetes node.
    sudo hostnamectl set-hostname "k8s-master01"
26. Reloads the shell to apply hostname changes.
    exec bash
27. Initializes the Kubernetes control-plane node.
    sudo kubeadm init --pod-network-cidr=10.244.0.0/16
28. Creates the Kubernetes configuration directory for kubectl.
    mkdir -p $HOME/.kube
29. Copies the admin kubeconfig file for cluster access.
    sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
30. Changes ownership of the kubeconfig file to the current user.
    sudo chown $(id -u):$(id -g) $HOME/.kube/config
31. Installs the Calico pod network plugin.
    kubectl apply -f https://raw.githubusercontent.com/projectcalico/calico/v3.29.1/manifests/calico.yaml
32. Enables the UFW firewall.
    sudo ufw enable
33. Allows SSH access through the firewall.
    sudo ufw allow ssh
34. Allows Kubernetes API server traffic through the firewall.
    sudo ufw allow 6443/tcp
35. Allows etcd communication ports.
    sudo ufw allow 2379:2380/tcp

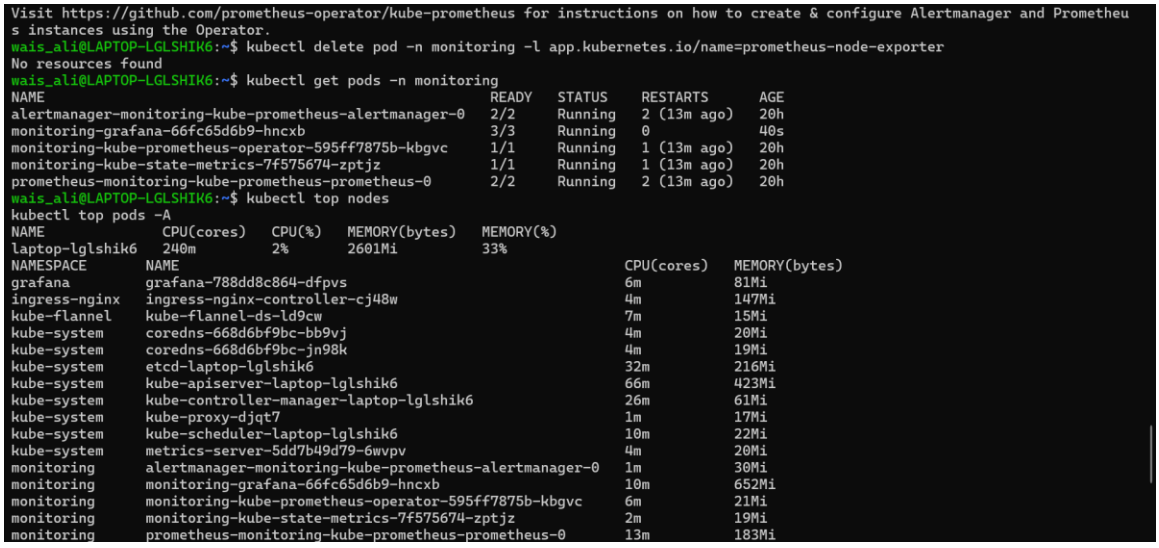
36. Allows kubelet API communication.
    sudo ufw allow 10250/tcp
37. Allows kube-controller-manager communication.
    sudo ufw allow 10257/tcp
38. Allows kube-scheduler communication.
    sudo ufw allow 10259/tcp
39. Allows Calico VXLAN networking traffic.
    sudo ufw allow 4789/udp
40. Allows Calico BGP traffic.
    sudo ufw allow 179/tcp
41. Allows pod network CIDR traffic.
    sudo ufw allow from 10.244.0.0/16
42. Removes control-plane taint to allow workloads on a single node.
    kubectl taint nodes --all node-role.kubernetes.io/control-plane-
43. Displays the status of Kubernetes nodes.
    kubectl get nodes
44. Lists all pods across all namespaces.
    kubectl get pods -A
45. Displays Kubernetes cluster information.
    kubectl cluster-info
46. Shows Kubernetes client and server versions.
    kubectl version
47. Displays the installed kubeadm version.
    kubeadm version

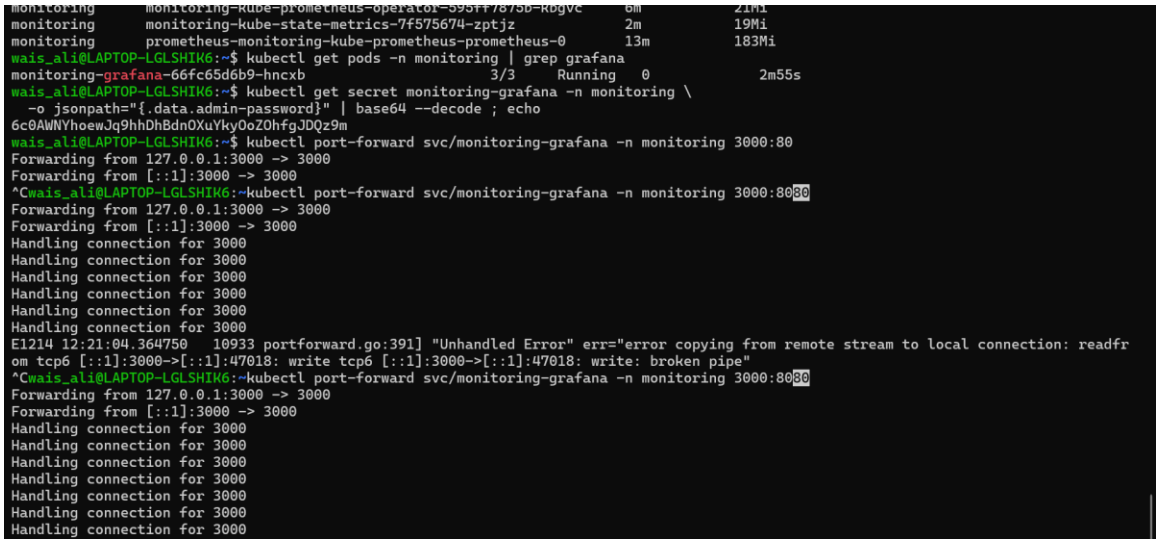


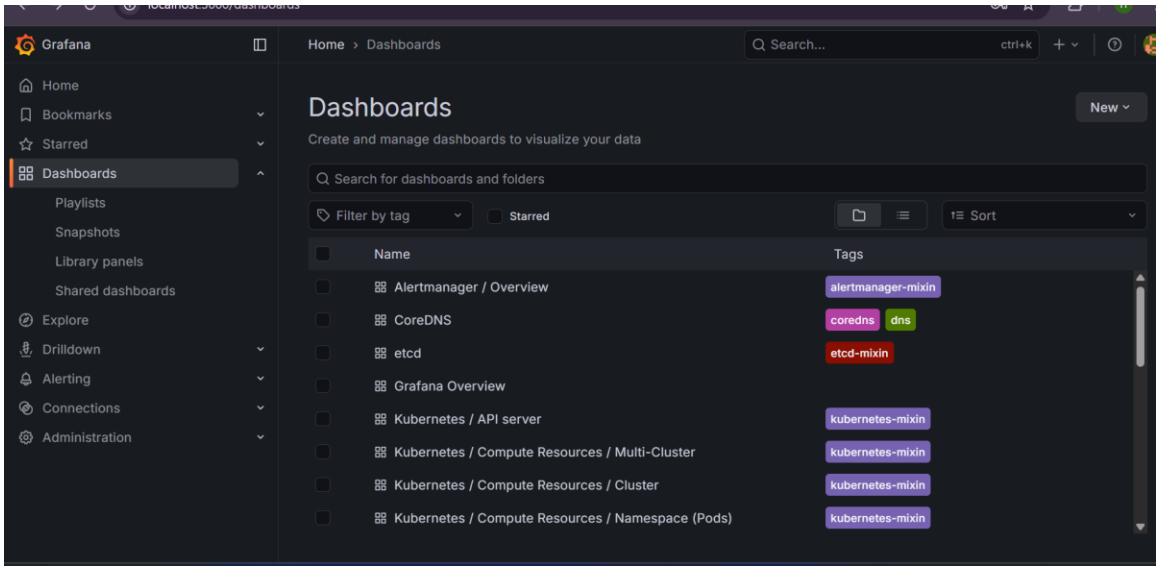
TASK#2

1. Downloads and installs Helm package manager.
   curl https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3 | bash
2. Verifies Helm installation and version.
   helm version
3. Adds the NGINX Ingress Controller Helm repository.
   helm repo add ingress-nginx https://kubernetes.github.io/ingress-nginx
4. Updates all Helm repositories.
   helm repo update
5. Installs NGINX Ingress Controller as a DaemonSet in the ingress-nginx namespace.
   helm install ingress-nginx ingress-nginx/ingress-nginx --namespace ingress-nginx --create-namespace --set controller.kind=DaemonSet --set controller.hostNetwork=false --set controller.service.type=ClusterIP
6. Displays the status of NGINX Ingress Controller pods.
   kubectl get pods -n ingress-nginx
7. Adds the Metrics Server Helm repository.
   helm repo add metrics-server https://kubernetes-sigs.github.io/metrics-server/
8. Updates Helm repositories after adding Metrics Server.
   helm repo update
9. Installs Metrics Server in the kube-system namespace.
   helm install metrics-server metrics-server/metrics-server -n kube-system --set args[0]=--kubelet-insecure-tls --set args[1]=--kubelet-preferred-address-types=InternalIP,Hostname
10. Displays CPU and memory usage of all nodes.
    kubectl top nodes
11. Displays CPU and memory usage of all pods in all namespaces.
    kubectl top pods -A
12. Adds the Prometheus Community Helm repository.
    helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
13. Updates Helm repositories after adding Prometheus Community.
    helm repo update
14. Installs Prometheus, Alertmanager, and Grafana monitoring stack.
    helm install monitoring prometheus-community/kube-prometheus-stack -n monitoring --create-namespace --set prometheusOperator.admissionWebhooks.enabled=false --set prometheusOperator.resources.requests.memory=200Mi --set prometheusOperator.resources.limits.memory=300Mi --set prometheus.resources.requests.memory=300Mi --set alertmanager.resources.requests.memory=200Mi --set nodeExporter.enabled=false

15. Displays the status of monitoring pods.
    kubectl get pods -n monitoring
16. Retrieves the Grafana admin password from Kubernetes secret.
    kubectl get secret monitoring-grafana -n monitoring -o jsonpath="{.data.admin-password}" | base64 --decode ; echo
17. Forwards Grafana service to local port 3000 for browser access.
    kubectl port-forward svc/monitoring-grafana -n monitoring 3000:80
18. Verifies monitoring pods are running correctly.
    kubectl get pods -n monitoring
19. Verifies node resource usage using Metrics Server.
    kubectl top nodes
20. Verifies pod resource usage across all namespaces.
    kubectl top pods -A







TASK#3

1. Deploys the Redis Leader deployment.
   kubectl apply -f https://k8s.io/examples/application/guestbook/redis-leader-deployment.yaml

2. Shows pods with app=redis and role=leader.
   kubectl get pods -l app=redis -l role=leader

3. Checks logs of the Redis Leader deployment (optional).
   kubectl logs -f deployment/redis-leader

4. Exposes Redis Leader via a Kubernetes Service.
   kubectl apply -f https://k8s.io/examples/application/guestbook/redis-leader-service.yaml

5. Displays the Redis Leader service.
   kubectl get service redis-leader

6. Deploys Redis Follower deployment.
   kubectl apply -f https://k8s.io/examples/application/guestbook/redis-follower-deployment.yaml

7. Shows pods with app=redis and role=follower.
   kubectl get pods -l app=redis -l role=follower

8. Exposes Redis Followers via a Kubernetes Service.
   kubectl apply -f https://k8s.io/examples/application/guestbook/redis-follower-service.yaml

9. Displays the Redis Follower service.
   kubectl get service redis-follower

10. Deploys Guestbook Frontend deployment.
    kubectl apply -f https://k8s.io/examples/application/guestbook/frontend-deployment.yaml

11. Shows pods with app=guestbook and tier=frontend.
    kubectl get pods -l app=guestbook -l tier=frontend

12. Exposes Guestbook Frontend via a Kubernetes Service.
    kubectl apply -f https://k8s.io/examples/application/guestbook/frontend-service.yaml

13. Displays the Frontend service.
    kubectl get service frontend

14. Port-forwards the frontend service to local port 8080.
    kubectl port-forward svc/frontend 8080:80

15. Scales the Frontend deployment up to 5 replicas.
    kubectl scale deployment frontend --replicas=5

16. Shows frontend pods after scaling.
    kubectl get pods -l app=guestbook -l tier=frontend

17. Scales the Frontend deployment down to 2 replicas.
    kubectl scale deployment frontend --replicas=2

18. Shows frontend pods after scaling down.
    kubectl get pods -l app=guestbook -l tier=frontend

19. Creates
    cat <<EOF | kubectl apply -f -
    apiVersion: networking.k8s.io/v1
    kind: Ingress
    metadata:
    name: guestbook-ingress
    namespace: default
    annotations:
    nginx.ingress.kubernetes.io/rewrite-target: /
    spec:
    rules:

- host: guestbook.local
  http:
  paths:

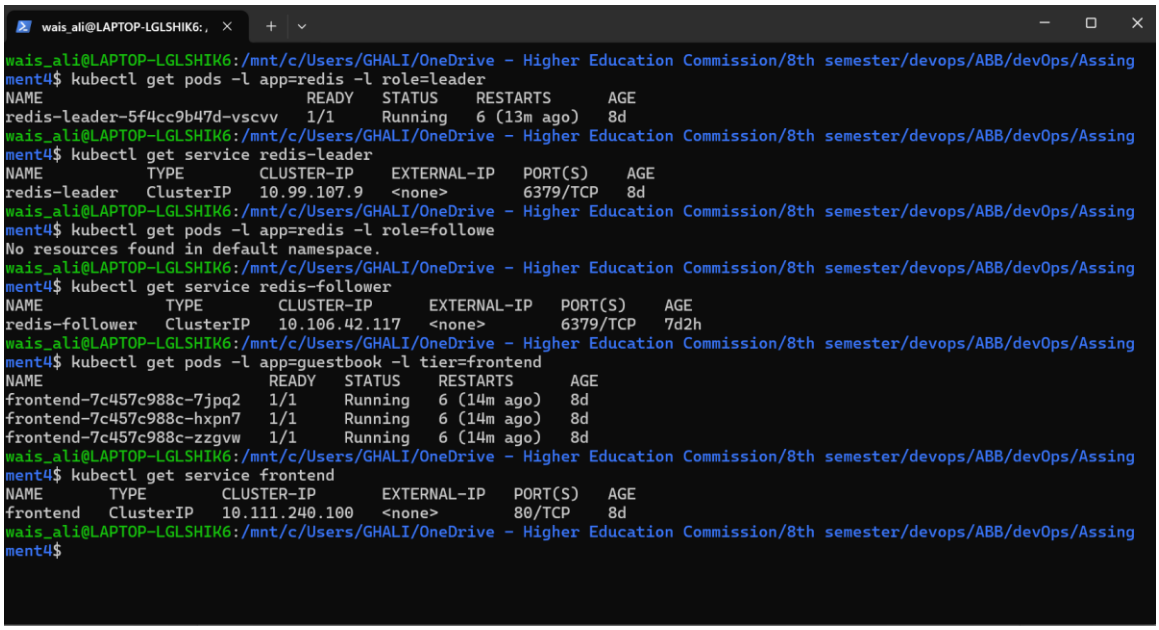
    - path: /
      pathType: Prefix
      backend:
      service:
      name: frontend
      port:
      number: 80
      EOF

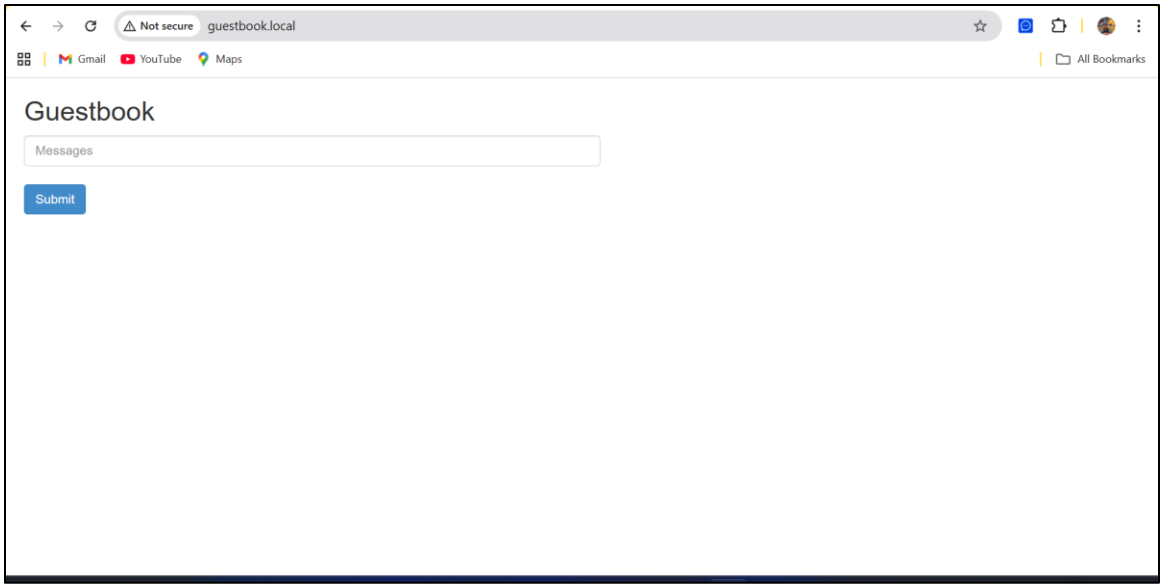
20. Displays the Guestbook ingress.
    kubectl get ingress guestbook-ingress

21. Adds host mapping for local testing (edit hosts file).
    sudo nano /etc/hosts

22. Adds line for Guestbook ingress to hosts file.
    127.0.0.1 guestbook.local

23. Tests Guestbook frontend in browser.
    http://guestbook.local





TASK#4

1. Removes existing Kubernetes APT repository.
   sudo rm -f /etc/apt/sources.list.d/kubernetes.list

2. Adds Kubernetes v1.33 APT repository.
   echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.33/deb/ /' | sudo tee /etc/apt/sources.list.d/kubernetes.list

3. Updates package index after adding new repository.
   sudo apt update

4. Unholds kubeadm package to allow upgrade on control plane.
   sudo apt-mark unhold kubeadm

5. Installs kubeadm v1.33 on control plane node.
   sudo apt install -y kubeadm=1.33.*

6. Holds kubeadm package to prevent automatic upgrades.
   sudo apt-mark hold kubeadm

7. Verifies installed kubeadm version.
   kubeadm version

8. Checks the Kubernetes upgrade plan.
   sudo kubeadm upgrade plan

9. Applies upgrade on control plane node (replace x with latest patch).
   sudo kubeadm upgrade apply v1.33.x

10. Unholds kubelet and kubectl to allow upgrade on control plane.
    sudo apt-mark unhold kubelet kubectl

11. Installs kubelet and kubectl v1.33 on control plane node.
    sudo apt install -y kubelet=1.33.* kubectl=1.33.*

12. Holds kubelet and kubectl to prevent automatic upgrades.
    sudo apt-mark hold kubelet kubectl

13. Reloads systemd manager configuration.
    sudo systemctl daemon-reload

14. Re-executes systemd manager.
    sudo systemctl daemon-reexec

15. Restarts kubelet service.
    sudo systemctl restart kubelet

16. Verifies Kubernetes client and server versions.
    kubectl version

17. Displays status of all nodes.
    kubectl get nodes

18. Removes Kubernetes APT repository on each worker node before upgrade.
    sudo rm -f /etc/apt/sources.list.d/kubernetes.list

19. Adds Kubernetes v1.33 APT repository on each worker node.
    echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.33/deb/ /' | sudo tee /etc/apt/sources.list.d/kubernetes.list

20. Updates package index on each worker node.
    sudo apt update

21. Unholds kubeadm package on worker node for upgrade.
    sudo apt-mark unhold kubeadm

22. Installs kubeadm v1.33 on worker node.
    sudo apt install -y kubeadm=1.33.*

23. Holds kubeadm to prevent automatic upgrades.
    sudo apt-mark hold kubeadm

24. Upgrades the worker node to match control plane version.
    sudo kubeadm upgrade node

25. Unholds kubelet and kubectl on worker node.
    sudo apt-mark unhold kubelet kubectl

26. Installs kubelet and kubectl v1.33 on worker node.
    sudo apt install -y kubelet=1.33.* kubectl=1.33.*

27. Holds kubelet and kubectl to prevent automatic upgrades.
    sudo apt-mark hold kubelet kubectl

28. Reloads systemd manager configuration on worker node.
    sudo systemctl daemon-reload

29. Re-executes systemd manager on worker node.
    sudo systemctl daemon-reexec

30. Restarts kubelet service on worker node.
    sudo systemctl restart kubelet

31. Verifies all nodes after upgrade.
    kubectl get nodes

32. Displays all pods across all namespaces to confirm cluster health.
    kubectl get pods -A

```
kubectl get pods -A
NAME             STATUS   ROLES           AGE   VERSION
laptop-lglshik6   Ready    control-plane   40h   v1.33.7
NAMESPACE       NAME                                                READY   STATUS    RESTARTS        AGE
default         frontend-7c457c988c-7jpq2                           1/1     Running   0               97m
default         frontend-7c457c988c-hxpn7                           1/1     Running   0               97m
default         frontend-7c457c988c-zzgvw                           1/1     Running   0               97m
default         redis-follower-854f6dcbd5-2k8hc                     1/1     Running   0               106m
default         redis-follower-854f6dcbd5-mv47z                     1/1     Running   0               106m
default         redis-leader-5f4cc9b47d-vscvv                       1/1     Running   0               125m
grafana         grafana-788dd8c864-dfpvs                            1/1     Running   1 (3h29m ago)   24h
ingress-nginx   ingress-nginx-controller-cj48w                      1/1     Running   2 (3h29m ago)   29h
kube-flannel    kube-flannel-ds-ld9cw                               1/1     Running   3 (3h29m ago)   40h
kube-system     coredns-674b8bbfcf-bkzx2                            1/1     Running   0               3m33s
kube-system     coredns-674b8bbfcf-bw6b6                            1/1     Running   0               3m33s
kube-system     etcd-laptop-lglshik6                                1/1     Running   1 (2m16s ago)   2m12s
kube-system     kube-apiserver-laptop-lglshik6                      1/1     Running   1 (2m16s ago)   2m12s
kube-system     kube-controller-manager-laptop-lglshik6             1/1     Running   1 (2m16s ago)   2m12s
kube-system     kube-proxy-47qv7                                    1/1     Running   0               3m33s
kube-system     kube-scheduler-laptop-lglshik6                      1/1     Running   0               2m12s
kube-system     metrics-server-5dd7b49d79-6wvpv                     1/1     Running   2 (3h29m ago)   29h
monitoring      alertmanager-monitoring-kube-prometheus-alertmanager-0   2/2     Running   2 (3h29m ago)   23h
monitoring      monitoring-grafana-66fc65d6b9-hncxb                 3/3     Running   0               3h15m
monitoring      monitoring-kube-prometheus-operator-595ff7875b-kbgvc   1/1     Running   1 (3h29m ago)   23h
monitoring      monitoring-kube-state-metrics-7f575674-zptjz        1/1     Running   3 (4m23s ago)   23h
monitoring      prometheus-monitoring-kube-prometheus-prometheus-0  2/2     Running   2 (3h29m ago)   23h
wais_ali@LAPTOP-LGLSHIK6:/mnt/c/Users/GHALI/OneDrive - Higher Education Commission/8th semester/devops/Assingment4/Task 3/aplication$
 kubectl version
Client Version: v1.33.7
Kustomize Version: v5.6.0
Server Version: v1.33.7
wais_ali@LAPTOP-LGLSHIK6:/mnt/c/Users/GHALI/OneDrive - Higher Education Commission/8th semester/devops/Assingment4/Task 3/aplication$
```